

The Transformer Architecture

Deep Learning

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Contents of This Video

In this video, we will cover:

- The “Attention is All You Need” paper
- Why Transformers replaced RNNs
- The encoder-decoder Transformer
- Key components overview
- The Transformer’s impact on AI

“Attention is All You Need” (2017)

The paper that changed everything:

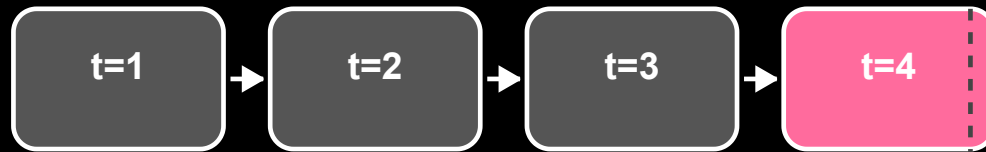
- Authors: Vaswani et al. (Google)
- Key insight: Replace recurrence entirely with attention
- Result: Better translation + 10x faster training

Why it mattered:

- RNNs process sequentially → slow
- Transformers process in parallel → fast
- Attention captures long-range dependencies directly

RNN vs. Transformer: Speed

RNN: $O(n)$ sequential steps



Sequential: Must wait for previous step

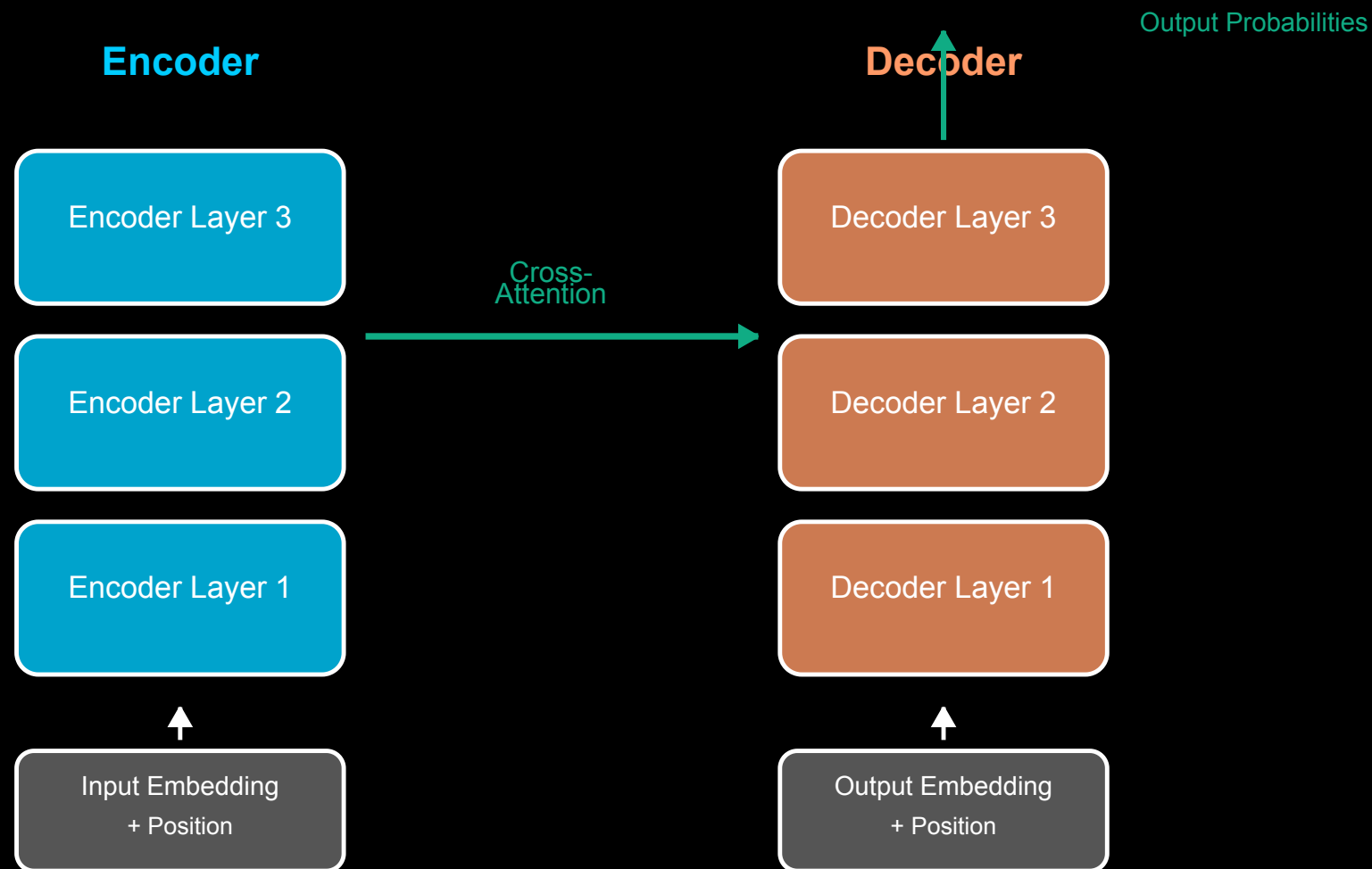
Transformer: $O(1)$ parallel steps



Parallel: All positions computed simultaneously

The Full Transformer Architecture

The Transformer: Encoder-Decoder Architecture



Key Components

Each encoder/decoder layer contains:

1. Multi-Head Self-Attention

- Every position attends to every position

2. Feed-Forward Network

- Two linear layers with ReLU
- Applied position-wise

3. Residual Connections

- Skip connections around each sublayer

4. Layer Normalization

- Stabilizes training

The Transformer's Impact

Before (2017): RNNs/LSTMs dominated NLP

After: Transformers everywhere

- **BERT** (2018): Understanding language
- **GPT** (2018-2023): Generating language
- **Vision Transformers** (2020): Images
- **ChatGPT** (2022): Conversational AI
- **GPT-4, Claude, etc.:** General AI assistants

Most important architecture in modern AI

What We've Covered

In this video, we've learned:

- “Attention is All You Need” (2017) introduced Transformers
- Transformers replace RNN recurrence with attention
- Parallel processing: much faster than sequential RNNs
- Encoder-decoder structure with stacked layers
- Components: self-attention, FFN, residuals, layer norm
- Foundation of BERT, GPT, and all modern language models